Shahjalal University of Science and Technology, Sylhet

Department of Electrical & Electronic Engineering

Course Code: EEE 0713 3237 Course Title: Power System II

Semester Final Examination (OBE): **Model Question**

OBE SET 1

Practice paper for self-study, mapped to the course learning outcomes (CO1–CO5). Not an official question paper.

Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

[Right margin: marks · CO = course outcome · BL = Bloom's level (C1 Remember, C2 Understand, C3 Apply, C4 Analyse, C5 Evaluate). Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

			Marks	CO	BL
1.	(a)	Define GMR and GMD. Why is r′ = 0.7788 r used in the inductance formula while the actual radius r is used for capacitance?	[2]	CO1	C2
	(b)	Show why the inductances of the three phases are unequal for unsymmetrical spacing, and derive the inductance per phase of a fully transposed three-phase line.	[4]	CO1	C3
	(c)	A 220 kV, 50 Hz, 150 km three-phase line uses a 2-conductor bundle per phase (sub-conductor diameter 3 cm, bundle spacing 40 cm). The phases are in a flat horizontal configuration 10 m apart (Fig. 1). Model the line by finding the inductance per phase (mH/km) and the total series reactance per phase.	[4]	CO1	C3

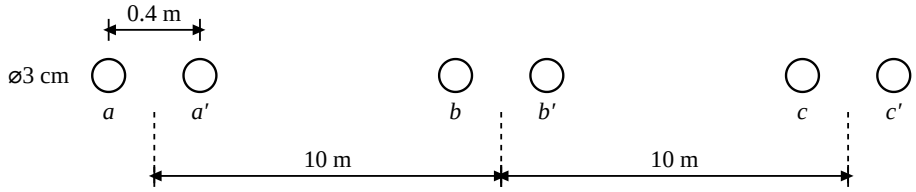


Fig. 1

2.	(a)	Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing.	[5]	CO1	C3
	(b)	For the line of Q1(c), find the capacitance to neutral (μF/km), the charging current per km and the total charging MVAr of the 150 km line.	[3]	CO1	C3
	(c)	Why must both L and C be included in the model of a long transmission line? How does the earth affect the line capacitance?	[2]	CO1	C2
3.	(a)	Compare overhead lines and underground cables on electrical, mechanical, environmental, safety and economic grounds. Which would you recommend for power distribution in a congested city like Dhaka? Justify.	[3]	CO2	C5
	(b)	Derive the expression for dielectric stress in a single-core cable and show that it is maximum at the conductor surface.	[3]	CO2	C3
	(c)	A single-core cable for a 66 kV, 3-phase system has conductor diameter 2.2 cm, sheath inside diameter 6 cm and $\varepsilon_r = 3.6$. Find g_{max}, g_{min}, capacitance per km and charging current per phase for a 5 km length. Is the design safe if the working stress of the insulation is 40 kV/cm (rms)?	[4]	CO2	C5

OR

			Marks	CO	BL
4.	(a)	Why is grading of cables needed? Explain intersheath grading with a figure.	[3]	CO2	C2
	(b)	The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is 4.5×10^{14} Ω-cm, find the insulation thickness.	[3]	CO2	C3
	(c)	Find the most economical conductor and sheath diameters of a single-core cable for a 220 kV, 3-phase system if the maximum permissible stress is 50 kV/cm (peak).	[2]	CO2	C3
	(d)	A 3 km, 11 kV, 3-phase belted cable has a capacitance of 1.8 μF/km between the three bunched cores and the sheath, and 1.0 μF/km between one core and the other two connected to the sheath. Find the charging current.	[2]	CO2	C3

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

			Marks	CO	BL
5.	(a)	Evaluate why suspension-type insulators are preferred to pin-type insulators above 33 kV.	[2]	CO2	C5
	(b)	Discuss the methods of improving string efficiency. Which method is most practical for EHV lines, and why?	[3]	CO2	C4
	(c)	A 132 kV, 3-phase line is supported by strings of 4 identical discs. The shunt capacitance of each link pin to earth is 1/9 of the self-capacitance of a disc. Analyse the voltage distribution across the discs and find the string efficiency. Which disc is most likely to fail first?	[5]	CO2	C4
6.	(a)	Explain the hazards created by corona (power loss, ozone, radio and telephone interference) and suggest mitigation methods.	[3]	CO4	C2
	(b)	A Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 Ω. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 Ω and the fixed resistors are equal. Calculate the distance of the fault from the test end.	[3]	CO4	C3
	(c)	A 2 km cable develops an earth fault. It is looped with an identical sound cable (0.8 Ω/km). In a Murray loop test the ratio arms at balance are 3.4 : 1. Draw the test circuit and locate the fault.	[4]	CO4	C3

OR

			Marks	CO	BL
7.	(a)	Compare HVDC and HVAC transmission with a cost–distance curve. Why does Bangladesh import power from India through the HVDC back-to-back station at Bheramara?	[4]	CO5	C5
	(b)	Explain series and shunt compensation. How does a series capacitor improve the power-transfer capability of a line?	[3]	CO3	C2
	(c)	A lossless line has X = 0.8 pu with $\lvert V_s \rvert = \lvert V_r \rvert = 1.0$ pu. Find the maximum power transfer. Find it again when 40 % series compensation is applied, and comment.	[3]	CO3	C4

8. (a) Distinguish between steady-state and transient stability. State the factors affecting transient stability. [3] CO4 C2

(b) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (|V| = 1.0 pu) with terminal voltage $|V_t|$ = 1.0 pu. Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred when (i) the system is healthy and (ii) one line is shorted (3-phase) in the middle. [4] CO4 C4

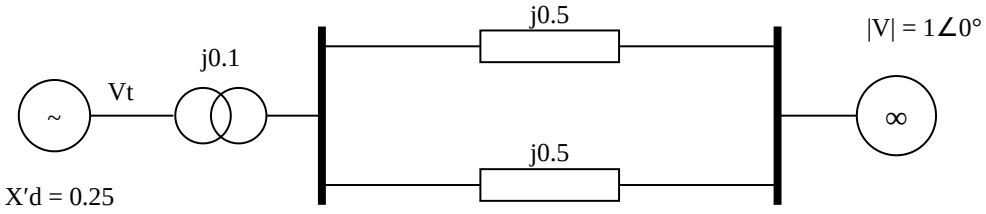


Fig. 2

(c) A 100 MVA, 50 Hz, 4-pole generator has H = 5 MJ/MVA. Find the kinetic energy stored at synchronous speed. If the mechanical input is suddenly raised by 25 MW with no change in electrical output, find the rotor acceleration in electrical degrees/s² and in rpm/s. [3] CO4 C3

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OBE SET 2

Practice paper for self-study, mapped to the course learning outcomes (CO1–CO5). Not an official question paper.

Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

[Right margin: marks · CO = course outcome · BL = Bloom's level (C1 Remember, C2 Understand, C3 Apply, C4 Analyse, C5 Evaluate). Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

			Marks	CO	BL
1.	(a)	What are skin effect and proximity effect? How do they influence the AC resistance of a line?	[2]	CO1	C2
	(b)	Derive the expression for the inductance of composite-conductor lines.	[5]	CO1	C3
	(c)	A single-phase 50 Hz power line (conductor spacing 2 m) carries 200 A. A telephone line (conductor spacing 1 m) runs symmetrically 3 m below it (Fig. 1). Find the mutual inductance and the voltage induced per km in the telephone line. Suggest one way to reduce this induced voltage.	[3]	CO1	C3

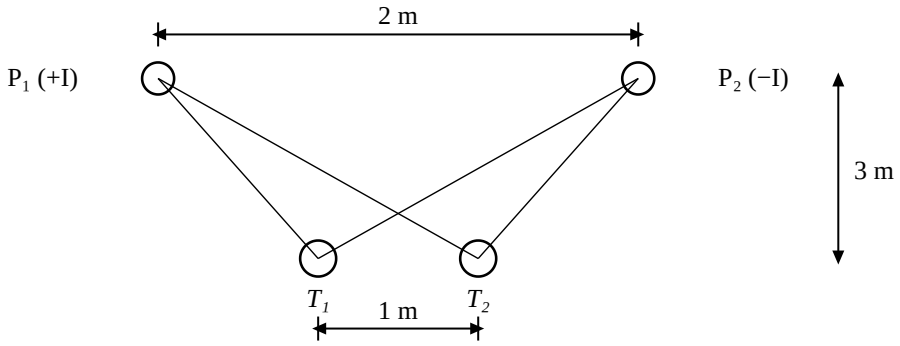


Fig. 1

2. (a) A 132 kV, 50 Hz, fully transposed three-phase line has its conductors at the corners of a triangle with sides 4 m, 5 m and 6 m (Fig. 2). Conductor diameter is 1.8 cm. Find the capacitance to neutral, the charging current per km and the charging MVAr of a 200 km length. [4] CO1 C3

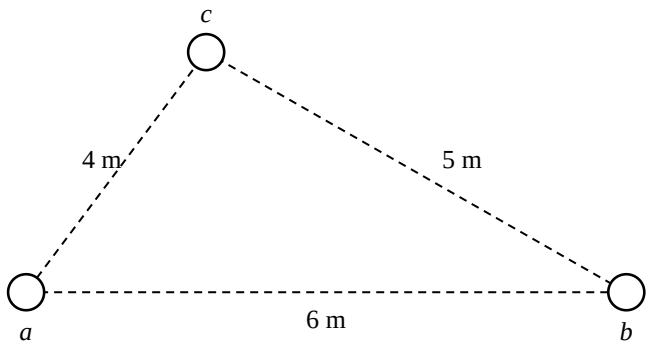


Fig. 2

(b) Derive the capacitance of a single-phase two-wire line and show that the capacitance to neutral is twice the line-to-line capacitance. [4] CO1 C3

(c) Using the method of images, explain why the presence of earth increases the line capacitance. When can this effect be neglected? [2] CO1 C2

3. (a) Write down the general construction of a 3-core cable with a neat figure. [3] CO2 C1

(b) Explain capacitance grading. For three dielectric layers, derive the expression for the potential difference between core and sheath. [4] CO2 C3

(c) A 33 kV single-core cable is graded with two dielectrics: $\varepsilon_r = 4$ (inner) and $\varepsilon_r = 2.5$ (outer). Core diameter 1.5 cm, the inner layer extends to 3 cm diameter, sheath inside diameter 5 cm. Find the maximum stress in each layer and compare it with an ungraded cable of the same dimensions. Evaluate the benefit of grading. [3] CO2 C5

OR

4. (a) Define thermal resistance of a cable. On what factors does the current rating of a cable depend? [2] CO2 C1

(b) A single-core cable laid in the ground has core diameter 25 mm and dielectric thickness 30 mm. The thermal resistivity of the dielectric is 5.5 thermal Ω-m and the thermal resistance between sheath and ground is 0.6 thermal Ω/m. If the temperature rise must not exceed 50 °C and R = 150 μΩ/m, estimate the permissible current. [4] CO2 C3

(c) A single-core cable for use on 11 kV, 50 Hz has a conductor area of 0.645 cm² and an internal sheath diameter of 2.18 cm. The relative permittivity of the dielectric is 3.5. Find (i) g_{max}, (ii) g_{min}, (iii) capacitance per km and (iv) charging current. [4] CO2 C3

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

			Marks	CO	BL
5.	(a)	Why is the voltage distribution over a suspension string unequal? For a 3-disc string, derive V_2 and V_3 in terms of V_1 and m.	[4]	CO2	C3
	(b)	A 66 kV, 3-phase line uses strings of 3 discs with m = 0.15. Find the voltage across each disc and the string efficiency.	[3]	CO2	C3
	(c)	A conductor (diameter 1.6 cm, weight 0.9 kg/m, ultimate strength 4000 kg) is strung on level supports 250 m apart with a safety factor of 2.5. Find the sag and the vertical sag when it is covered with 1 cm of radial ice (density 915 kg/m³) and carries a wind pressure of 40 kg/m² of projected area.	[3]	CO2	C4
6.	(a)	What are the causes of insulation failure? How does electrical breakdown happen in a pin-type insulator?	[3]	CO4	C2
	(b)	A 220 kV, 50 Hz, 3-phase line has conductors of 2.5 cm diameter at 4 m equilateral spacing. At 25 °C and 76 cm Hg, with irregularity factor 0.85 and g_0 = 21.1 kV/cm (rms), find the critical disruptive voltage. Will corona occur in fair weather? Comment.	[4]	CO4	C4
	(c)	How can an earth fault in an underground cable become a safety hazard? Explain how a Megger is used to identify open-circuit, short-circuit and earth faults.	[3]	CO4	C2
		OR			
7.	(a)	Compare SVC, STATCOM, TCSC and UPFC in terms of connection, operating principle and application.	[4]	CO3	C4
	(b)	Describe the types of HVDC links. Which link would you choose for a long submarine connection, and why?	[3]	CO5	C5
	(c)	Define voltage sag, swell, harmonics and flicker. How do they affect sensitive loads?	[3]	CO3	C2
8.	(a)	Derive the equal-area criterion for a single machine connected to an infinite bus.	[4]	CO4	C3
	(b)	Using the P–δ curve, explain what happens for a sudden increase in mechanical input to a generator.	[3]	CO4	C2
	(c)	A generator (E = 1.2 pu) delivers 0.8 pu to an infinite bus (V = 1.0 pu) through a total reactance of 0.6 pu. The mechanical input is suddenly increased to 1.2 pu. Using the equal-area criterion, find δ_0, the new steady angle δ_1 and the maximum swing δ_2. Is the system stable?	[3]	CO4	C4

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Semester Final Examination (OBE): **Model Question**

OBE SET 3

Practice paper for self-study, mapped to the course learning outcomes (CO1–CO5). Not an official question paper.

Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

[Right margin: marks · CO = course outcome · BL = Bloom's level (C1 Remember, C2 Understand, C3 Apply, C4 Analyse, C5 Evaluate). Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

			Marks	CO	BL
1.	(a)	What is transposition of a transmission line? Why is it used?	[2]	CO1	C2
	(b)	Derive the expression for the flux linkages of one conductor in a group of conductors.	[4]	CO1	C3
	(c)	A 400 kV, 50 Hz line uses a 3-conductor triangular bundle (side 0.4 m, sub-conductor GMR 1.2 cm). The phases are flat, 12 m apart (Fig. 1). Find the inductance and reactance per km, and analyse the percentage reduction of inductance compared with a single conductor of the same GMR.	[4]	CO1	C4

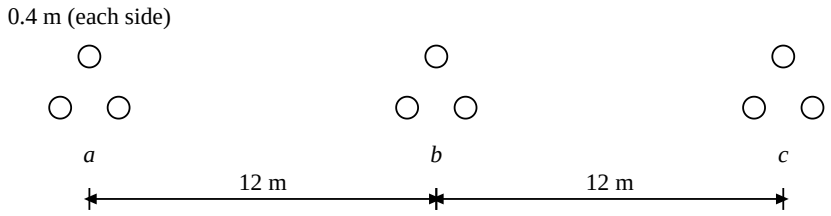


Fig. 1

2.	(a)	Derive the expression for the capacitance of a three-phase line with equilateral spacing.	[3]	CO1	C3
	(b)	A three-phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard-drawn seven-strand copper (outside diameter = 1.05 cm). The line voltage is 110 kV. Find the capacitance to neutral and the charging current per km of line.	[3]	CO1	C3
	(c)	For the line of Q1(c) (sub-conductor radius 1.5 cm), find the capacitance to neutral, the charging current per km and the total charging MVAr for 300 km. Why do such EHV lines need shunt reactors at light load?	[4]	CO1	C5

3. (a) Derive the condition for the most economical conductor size in a cable. [3] CO2 C3

(b) A 33 kV, 3-phase, 5 km cable system uses three single-core cables with conductor diameter 2 cm, insulation thickness 0.6 cm, $\rho = 6 \times 10^{12}$ Ω-m and $\varepsilon_r = 3.2$. Find the insulation resistance, the capacitance per phase, the charging current and the total charging kVAR. [4] CO2 C3

(c) A 400 kV AC submarine cable (C = 0.2 μF/km) is proposed for a 60 km sea crossing. Calculate its charging current and evaluate whether AC or HVDC should be chosen. [3] CO5 C5

OR

4. (a) Classify cables by voltage. Why are belted cables limited to about 11 kV, and how do H-type cables overcome this? [3] CO2 C2

(b) A 66 kV single-core cable (core diameter 1.5 cm, sheath inside diameter 5 cm) has one intersheath at 3 cm diameter. Find the voltage that must be applied between core and intersheath for equal maximum stress, and compare g_{max} with and without the intersheath. [4] CO2 C4

(c) The maximum and minimum stresses in the dielectric of a single-core cable are 40 kV/cm (rms) and 10 kV/cm (rms) respectively. If the conductor diameter is 2 cm, find (i) the thickness of insulation and (ii) the operating voltage. [3] CO2 C3

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

		Marks	CO	BL

5. (a) Name the types of insulators with their voltage ranges. Select a suitable type for (i) an 11 kV feeder, (ii) a 132 kV line and (iii) a dead-end tower. Justify. [3] CO2 C5

(b) In a string of 4 identical discs, the voltages across the top (tower-side) unit and the second unit are 9 kV and 11 kV. Find m, the voltages across the remaining discs, the line voltage and the string efficiency. [4] CO2 C4

(c) Explain the static shielding (guard ring) method of improving string efficiency with a figure. [3] CO2 C2

6. (a) Two towers of heights 40 m and 55 m support a conductor over level ground. The span is 350 m, the conductor weighs 0.85 kg/m and the tension is 1800 kg. Find the position of the lowest point and its clearance above the ground. [4] CO2 C3

(b) Define critical disruptive voltage. Write down its significance. [2] CO2 C1

(c) Why does sag increase in summer? What hazards arise from insufficient ground clearance, and how is a stringing chart used to avoid them? [4] CO4 C4

OR

7. (a) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistances of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end. [4] CO4 C3

(b) A Varley loop test is made on a 15 km cable of 25 Ω/km, looped with an identical healthy conductor. With equal ratio arms, balance is obtained when the variable resistance is 150 Ω. Locate the fault. [3] CO4 C3

(c) Classify the types of cable faults and explain how each one is detected. [3] CO4 C2

8. (a) Derive the swing equation of two coherent machines and find H_{eq} on a 100 MVA base for machines of 500 MVA (H = 4 MJ/MVA) and 300 MVA (H = 6 MJ/MVA). [3] CO4 C3

(b) In Fig. 2, E′ = 1.2 pu and |V| = 1.0 pu. The generator delivers 1.0 pu. Find P_{max} and δ_0 for the healthy system. If one line trips, find the new P_{max} and operating angle and analyse whether steady-state operation is still possible. [4] CO4 C4

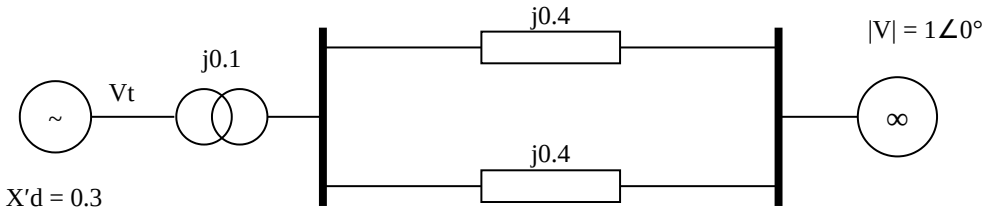


Fig. 2

(c) State the factors affecting stability. How do modern technologies (fast breakers, auto-reclosing, FACTS, HVDC) improve the stability of a power system? [3] CO3 C2

CO Map, Answer Key & Previous-Year Share

Course learning outcomes (from 05_Course_Information/course_learning_outcomes.pdf)

CO	Statement	Where it is examined in these sets
CO1	Formulate mathematical models using the effect of inductance and capacitance for interconnected power networks.	Ch-2 & Ch-3: L, C derivations, GMD/GMR, bundles, mutual inductance, charging current
CO2	Evaluate the suitability of overhead and underground transmission considering electrical, mechanical, environmental, performance, safety and economic constraints.	Ch-8 & Ch-11: cables (stress, grading, rating), insulators, string efficiency, sag, OH vs UG choice
CO3	Explain and critically evaluate technologies and information systems for operational management of conventional and modern power systems.	Compensation, FACTS, power quality
CO4	Apply basic principles to formulate significant electrical hazards; analyse stability by measuring appropriate parameters.	Cable faults & loop tests, corona hazards, clearance, insulation failure; Ch-12 stability
CO5	Follow advancements in technology; choose the appropriate transmission system considering cost, environment, society and ethics.	HVDC vs HVAC, AC cable vs HVDC, OH vs UG in cities, modern stability aids

Marks per CO (all 8 questions = 80 printed marks) and previous-year (PYQ) share

Set	CO1	CO2	CO3	CO4	CO5	Total	PYQ marks in any answer path
OBE Set 1	20	30	6	20	4	80	9–15 of 60 (15–25 %)
OBE Set 2	20	30	7	20	3	80	11–15 of 60 (18–25 %)
OBE Set 3	20	33	3	21	3	80	10–12 of 60 (17–20 %)

PYQ = repeated word-for-word (same numbers) from the 2021–2024 finals or term tests. Target was about 20 % of 60 marks, as you said the teacher does.

- **Set 1:** 2(a), 4(b), 6(b), 8(b)
- **Set 2:** 1(b), 3(a), 4(c), 6(a), 8(b)
- **Set 3:** 1(a), 2(b), 3(a), 4(c), 6(b), 7(a)

OBE Set 1: final answers

- 1(c) D_s = 0.0683 m, D_{eq} = 12.6 m; L = 1.043 mH/km; X = 0.328 Ω/km → 49.2 Ω for 150 km
- 2(b) $r_{eq} = \sqrt{(rd)}$ = 0.0775 m; C_n = 0.01093 µF/km; I_c = 0.436 A/km; 24.9 MVAr
- 3(c) V_{ph} = 38.1 kV; g_{max} = 34.5 kV/cm, g_{min} = 12.7 kV/cm; C = 0.200 µF/km; I_c = 11.9 A; safe (34.5 < 40)
- 4(b) ≈ 1.25 cm [PYQ] · 4(c) d = 7.19 cm, D = 19.5 cm · 4(d) C_e = 0.6, C_c = 0.2 µF/km; C_N = 3.6 µF; I_c = 7.18 A
- 5(c) V = 14.69, 16.33, 19.77, 25.42 kV (bottom disc fails first); η = 75.0 %
- 6(b) 15 km [PYQ] · 6(c) X = 3.2/4.4 = 0.727 Ω → 0.909 km from test end
- 7(c) P_{max} = 1.25 pu → 2.08 pu with 40 % compensation (+67 %)
- 8(b) $E' = 1.075\angle 33.9°$; P_{max} = 1.79 / 0.693 pu [PYQ] · 8(c) KE = 500 MJ; α = 450 elec °/s² = 37.5 rpm/s

OBE Set 2: final answers

- 1(c) M = 0.0391 mH/km; V = 2.46 V/km; reduce by transposing the telephone line / more spacing
- 2(a) D_{eq} = 4.93 m; C_n = 0.00882 µF/km; I_c = 0.211 A/km; 9.66 MVAr
- 3(c) g_{1max} = 29.1, g_{2max} = 23.3 kV/cm vs ungraded 36.5 kV/cm (about 20 % lower)
- 4(b) S = 1.671 thermal Ω/m; I ≈ 447 A · 4(c) 27.66 / 11.5 kV/cm, 0.222 µF/km, 0.766 A [PYQ]
- 5(b) 10.52, 12.10, 15.49 kV; η = 82.0 % · 5(c) w_{ice} = 0.747, w_{wind} = 1.44, w_t = 2.188 kg/m; S = 10.68 m, vertical 8.04 m
- 6(b) δ = 1.0; V_c = 129.3 kV/ph (223.9 kV line) > 220 kV operating, so no corona in fair weather but the margin is small (corona likely in bad weather)
- 8(c) δ_0 = 23.6°, δ_1 = 36.9°, δ_2 = 51.0° < δ_{max} = 143.1° → stable

OBE Set 3: final answers

- 1(c) D_s^b = 0.1243 m, D_{eq} = 15.12 m; L = 0.960 mH/km, X = 0.302 Ω/km; 32.7 % less than a single conductor (1.428 mH/km)
- 2(b) 0.00826 µF/km, 0.165 A/km [PYQ] · 2(c) r_{eq} = 0.134 m; C_n = 0.01177 µF/km; I_c = 0.854 A/km; 177.5 MVAr → shunt reactors to absorb VAR / control the Ferranti rise
- 3(b) R = 89.8 MΩ; C = 1.89 µF/phase; I_c = 11.33 A; 648 kVAR · 3(c) I_c ≈ 871 A/phase (≈ 603 MVAr): the cable's thermal capacity goes on charging current → choose HVDC
- 4(b) $V_{core–intersheath}$ = 26.7 kV (intersheath at 39.3 kV); g_{max} = 51.3 vs 73.1 kV/cm without · 4(c) 3 cm, 55.45 kV [PYQ]
- 5(b) m = 0.222; V = 9, 11, 15.44, 23.32 kV; V_{ph} = 58.77 kV, V_L = 101.8 kV; η = 63.0 %
- 6(a) x_1 = 84.2 m from the 40 m tower; S_1 = 1.68 m → lowest point 38.3 m above ground
- 7(a) 433.3 m [PYQ] · 7(b) X = (750 − 150)/2 = 300 Ω → 12 km
- 8(a) H_{eq} = 38 MJ/MVA (100 MVA base) · 8(b) healthy X = 0.6 → P_{max} = 2.0, δ_0 = 30°; one line out X = 0.8 → P_{max} = 1.5, δ = 41.8° → still stable (steady state)

How to use: sit each set as a 2-hour mock. The PYQ parts are free marks, so learn them perfectly (solutions are in 03_Solved_Math_Bank.md). The new CO-based parts test the same formulas with new numbers and "evaluate / justify / compare" wording. Practise writing a short justification (2–3 lines) at the end of every numerical, because OBE marking rewards it.